



Week 2 Lecture 1

▼ Class	BSCCS2003
🕒 Created	@September 14, 2021 12:16 PM
🔗 Materials	
# Module #	10
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Information Representation in a Machine

Markup

- Information representation
- Raw data vs Semantics
- Logical structure vs Styling
- HTML5 & CSS

Information representation

- Computers work only with "bits"
 - Binary digits: 0 and 1
- Numbers
 - Place value: binary numbers: eg. 6 = 0110
 - Two's complement: negative numbers: eg. -6 = 1010

Representing Text

- ASCII
- Unicode
- UTF-8

Information Interchange

- Communicate through machines - either between machines or between humans
- Machines only work with **bits**
- Standard "encoding"
 - Some sequence of bits interpreted as a character

Interpretation

What is "0100 0001"?

- It is a string of bits
- Number with a value of 65 in decimal
- Character "A"
- All of the above statements are correct

It boils down to a matter of **interpretation** and **context**

ASCII

American Standard Code for Information Interchange

- 7-bits: 128 different entities
 - 'a'..'z'
 - 'A'..'Z'
 - '0'..'9'
 - Special characters- ! @ # \$ % ^ & * () ...
- Why 7-bits?

It was a time before even 8-bit computers were standardised

The number of bits were precious and you really didn't want to use up more than necessary bits

That is why people tried to optimise the number of bits

- What about other characters? あ आ 아 𐄌
 - 1000s of characters needed

Unicode

- Allow codes for more script, characters
- How many?
 - All the living languages?
 - All extinct languages?
 - All future languages?
- **"Universal Character Set"** encoding - UCS
 - **UCS-2** 2 bytes per character - max 65,536 characters
 - **UCS-4** 4 bytes per character - 4B+ characters